

Linear Algebra

Autonomous Navigation of Mobile Robots

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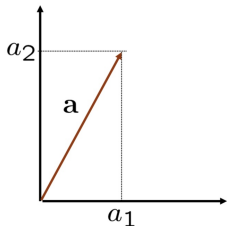
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Vectors

- Arrays of numbers
- Vectors represent a point in a n dimensional space

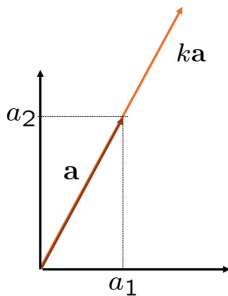
$$(a_1), \quad \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}, \quad \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$



Vectors

Scalar Product

- Scalar-Vector Product $k\mathbf{a}$
- Changes the length of the vector, but not its direction



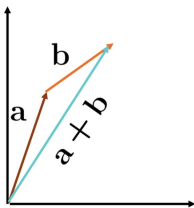
Vectors

Sum

- Sum of vectors (is commutative)

$$\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} + \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

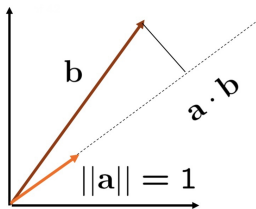
- Can be visualized as “chaining” the vectors



Vectors

Dot Product

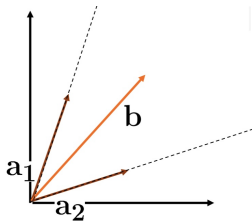
- Inner product of vectors (is a scalar)
- If one of the two vectors, e.g. $\mathbf{a}$, has $\|\mathbf{a}\| = 1$, the inner product $\mathbf{a} \cdot \mathbf{b}$ returns the length of the projection of $\mathbf{b}$ along the direction of $\mathbf{a}$
- If $\mathbf{a} \cdot \mathbf{b} = 0$, the two vectors are **orthogonal**



Vectors

Linear (In)Dependence

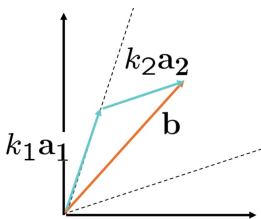
- A vector **b** is **linearly dependent** from $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$ if
$$\mathbf{b} = \sum_i k_i \mathbf{a}_i$$
- In other words, if **b** can be obtained by summing up the $\mathbf{a}_i$ properly scaled
- If there exist no $\{k_i\}$ such that $\mathbf{b} = \sum_i k_i \mathbf{a}_i$ then **b** is independent from $\mathbf{a}_i$



Vectors

Linear (In)Dependence

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Matrices

- A matrix is written as a table of values

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & & & \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix}, \quad A : m \times n$$

- 1st index refers to the **row**
- 2nd index refers to the **column**
- Note: a d -dimensional vector is equivalent to a $d \times 1$ matrix

Matrices

as Collections of Vectors

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & & & \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix}$$

■ Column vectors

$$A = (a_{*1} \quad a_{*2} \quad \cdots \quad a_{*m})$$

Matrices

as Collections of Vectors

■ Row vectors

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & & & \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix} = \begin{pmatrix} a_{1*}^T \\ a_{2*}^T \\ \vdots \\ a_{n*}^T \end{pmatrix}$$

Important Matrices Operations

- Multiplication by a scalar
- Sum (commutative, associative)
- Multiplication by a vector
- Product (not commutative)
- Inversion (square, full rank)
- Transposition

Scalar Multiplication & Sum

- In the scalar multiplication, every element of the vector or matrix is multiplied with the scalar
- The sum of two vectors is a vector consisting of the pair-wise sums of the individual entries
- The sum of two matrices is a matrix consisting of the pair-wise sums of the individual entries

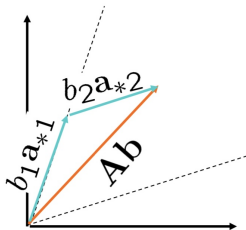
Matrix Vector Product

- The i^{th} component of $\mathbf{Ab}$ is the dot product $\mathbf{a}_{i*}^T \cdot \mathbf{b}$
- The vector $\mathbf{Ab}$ is linearly dependent from the column vectors $\{\mathbf{a}_{*i}\}$ with coefficients $\{b_i\}$

$$\mathbf{Ab} = \begin{pmatrix} \mathbf{a}_{1*}^T \\ \mathbf{a}_{2*}^T \\ \vdots \\ \mathbf{a}_{n*}^T \end{pmatrix} \cdot \mathbf{b} = \begin{pmatrix} \mathbf{a}_{1*}^T \cdot \mathbf{b} \\ \mathbf{a}_{2*}^T \cdot \mathbf{b} \\ \vdots \\ \mathbf{a}_{n*}^T \cdot \mathbf{b} \end{pmatrix} = \sum_k \mathbf{a}_{*k} b_k$$

Matrix Vector Product

- If the column vectors of $\mathbf{A}$ represent a reference system, the product $\mathbf{A}\mathbf{b}$ computes the global transformation of the vector $\mathbf{b}$ according to $\{\mathbf{a}_{*j}\}$



Matrix Matrix Product

Can be defined through

- the dot product of row and column vectors
- the linear combination of the columns of **A** scaled by the coefficients of the columns of **B**

$$\begin{aligned} \mathbf{C} = \mathbf{AB} = & \\ & \begin{pmatrix} \mathbf{a}_{1*}^T \cdot \mathbf{b}_{*1} & \mathbf{a}_{1*}^T \cdot \mathbf{b}_{*2} & \cdots & \mathbf{a}_{1*}^T \cdot \mathbf{b}_{*m} \\ \mathbf{a}_{2*}^T \cdot \mathbf{b}_{*1} & \mathbf{a}_{2*}^T \cdot \mathbf{b}_{*2} & \cdots & \mathbf{a}_{2*}^T \cdot \mathbf{b}_{*m} \\ \vdots & & & \\ \mathbf{a}_{n*}^T \cdot \mathbf{b}_{*1} & \mathbf{a}_{n*}^T \cdot \mathbf{b}_{*2} & \cdots & \mathbf{a}_{n*}^T \cdot \mathbf{b}_{*m} \end{pmatrix} = \\ & (\mathbf{A} \cdot \mathbf{b}_{*1} \quad \mathbf{A} \cdot \mathbf{b}_{*2} \quad \cdots \quad \mathbf{A} \cdot \mathbf{b}_{*m}) \end{aligned}$$

Matrix Matrix Product

Can be defined through

- If we consider the second interpretation, we see that the columns of $\mathbf{C}$ are the “transformations” of the columns of $\mathbf{B}$ through $\mathbf{A}$
- All the interpretations made for the matrix vector product hold

$$\mathbf{C} = \mathbf{AB} = (\mathbf{A} \cdot \mathbf{b}_{*1} \quad \mathbf{A} \cdot \mathbf{b}_{*2} \quad \cdots \quad \mathbf{A} \cdot \mathbf{b}_{*m})$$
$$\mathbf{c}_{*j} = \mathbf{A}\mathbf{b}_{*j}$$

Rank

- **Maximum** number of linearly independent rows (columns)
- Dimension of the **image** of the transformation $f(\mathbf{x}) = \mathbf{Ax}$
- When $\mathbf{A}$ is $m \times n$ we have
 - $\text{rank}(\mathbf{A}) \geq 0$ and the equality holds iff $\mathbf{A}$ is the null matrix
 - $\text{rank}(\mathbf{A}) \leq \min(m, n)$
- Computation of the rank is done by
 - Gaussian elimination on the matrix
 - Counting the number of non-zero rows

Inverse

$$\mathbf{AB} = \mathbf{I}$$

- If $\mathbf{A}$ is a square matrix of full rank, then there is a unique matrix $\mathbf{B} = \mathbf{A}^{-1}$ such that $\mathbf{AB} = \mathbf{I}$ holds
- The i^{th} row of $\mathbf{A}$ is and the j^{th} column of $\mathbf{A}^{-1}$ are:
 - orthogonal (if $i \neq j$)
 - or their dot product is 1 (if $i = j$)

Matrix Inversion

$$\mathbf{A}\mathbf{B} = \mathbf{I}$$

- The i^{th} column of $\mathbf{A}^{-1}$ can be found by solving the following linear system:

$$\mathbf{A}\mathbf{a}_{*i}^{-1} = \mathbf{i}_{*i}$$

where $\mathbf{i}_{*i}$ is the i^{th} column of the identity matrix

Determinant

- Only defined for **square matrices**
- The inverse of **A** exists if and only if $\det(\mathbf{A}) \neq 0$
- For 2×2 matrices: Let $\mathbf{A} = [a_{ij}]$ and $|\mathbf{A}| = \det(\mathbf{A})$, then

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

- For 3×3 matrices the Sarrus rule holds:

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} =$$
$$a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} -$$
$$a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}$$

Determinant

- For **general** $n \times n$ matrices?

Let $\mathbf{A}_{ij}$ be the submatrix obtained from by deleting the i^{th} row and the j^{th} column

$$\begin{bmatrix} 1 & 2 & 5 & 0 \\ 2 & 3 & 4 & -1 \\ -5 & 8 & 0 & 0 \\ 0 & 4 & -2 & 0 \end{bmatrix} \rightarrow \mathbf{A}_{32} = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

Rewrite determinant for 3×3 matrices:

$$\begin{aligned} \det(\mathbf{A}^{3 \times 3}) &= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - \\ &\quad a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{11} = \\ &= a_{11} \cdot \det(\mathbf{A}_{11}) - a_{12} \cdot \det(\mathbf{A}_{12}) + a_{13} \cdot \det(\mathbf{A}_{13}) \end{aligned}$$

Determinant

- For **general** $n \times n$ matrices?

$$\det(\mathbf{A}) = a_{11}\det(\mathbf{A}_{11}) - a_{12}\det(\mathbf{A}_{12}) + \dots + (-1)^{1+n}a_{1n}\det(\mathbf{A}_{1n}) = \sum_{j=1}^n (-1)^{1+j}a_{1j}\det(\mathbf{A}_{1j})$$

Let $\mathbf{C}_{ij} = (-1)^{1+j}\det(\mathbf{A}_{1j})$ be the (i, j) -cofactor, then

$$\det(\mathbf{A}) = a_{11}\mathbf{C}_{11} + a_{12}\mathbf{C}_{12} + \dots + a_{1n}\mathbf{C}_{1n} = \sum_{j=1}^n a_{1j}\mathbf{C}_{1j}$$

This is called the **cofactor expansion** across the first row

Determinant

- **Problem:** Take a 25×25 matrix (which is considered small). The cofactor expansion method requires $n!$ multiplications. For $n = 25$, this is 1.5×10^{25} multiplications for which would take **500,000 years**
- There are **much faster** methods, namely using **Gauss elimination** to bring the matrix into triangular form

$$\mathbf{A} = \begin{bmatrix} d_1 & * & * & * \\ 0 & d_2 & * & * \\ 0 & 0 & d_3 & * \\ 0 & 0 & 0 & d_4 \end{bmatrix} \quad \det(\mathbf{A}) =$$

Because for **triangular matrices** the determinant is the product of diagonal elements

Determinant

Properties

- **Row operations** (**A** is still a square matrix)
 - If **B** results from **A** by interchanging two rows, then $\det(\mathbf{B}) = -\det(\mathbf{A})$
 - If **B** results from **A** by multiplying one row with a number c , then $\det(\mathbf{B}) = c \cdot \det(\mathbf{A})$
 - If **B** results from **A** by adding a multiple of one row to another row, then $\det(\mathbf{B}) = \det(\mathbf{A})$
- **Transpose** $\det(\mathbf{A}^T) = \det(\mathbf{A})$
- **Multiplication** $\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$
- Does **not** apply to addition! $\det(\mathbf{A} + \mathbf{B}) \neq \det(\mathbf{A}) + \det(\mathbf{B})$

Determinant

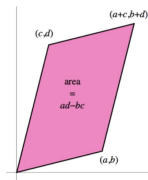
Properties

■ Compute Eigenvalues

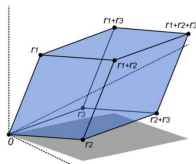
Solve the characteristic polynomial $\det(\mathbf{A} - \lambda \cdot \mathbf{I}) = 0$

■ Area and Volume

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$



$$\mathbf{A} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$



r is i^{th} row

Orthogonal Matrix

- A matrix $\mathbf{Q}$ is **orthogonal** iff its column (row) vectors represent an **orthonormal** basis

$$\mathbf{q}_{*i}^T \cdot \mathbf{q}_{*j} = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases} \quad \forall i, j$$

- Some properties
 - The transpose is the inverse

$$\mathbf{Q}\mathbf{Q}^T = \mathbf{Q}^T\mathbf{Q} = \mathbf{I}$$

- Determinant has unity norm (± 1)

$$1 = \det(\mathbf{I}) = \det(\mathbf{Q}^T\mathbf{Q}) = \det(\mathbf{Q})\det(\mathbf{Q}^T) = \det(\mathbf{Q})^2$$

Rotation Matrix

- A Rotation matrix is an orthonormal matrix with $\det = +1$

- 2D Rotations

$$\mathbf{R}(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

- 3D Rotations

$$\mathbf{R}_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{bmatrix} \quad \mathbf{R}_y(\theta) = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix}$$

- Rotations are not commutative

Matrices to Represent Affine Transformations

- A general and easy way to describe a 3D transformation is via matrices

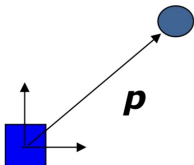
$$\mathbf{A} = \begin{pmatrix} \mathbf{R} & \mathbf{t} \\ 0 & 1 \end{pmatrix}, \quad \mathbf{A}^{-1} = \begin{pmatrix} \mathbf{R}^T & -\mathbf{R}^T \mathbf{t} \\ 0 & 1 \end{pmatrix}, \quad \mathbf{p} = \begin{pmatrix} \mathbf{t} \\ 1 \end{pmatrix}$$

- Takes naturally into account the non-commutativity of the transformations
- Homogeneous coordinates

Combining Transformations

A simple interpretation: chaining of transformations (represented as homogeneous matrices)

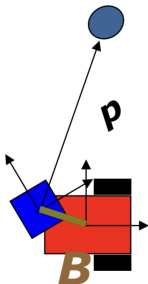
- Matrix **A** represents the pose of a **robot** in the space
- Matrix **B** represents the position of a sensor on the robot
- The **sensor** perceives an **object** at a given location **p**, in its own frame (the sensor has no clue on where it is in the world)
- Where is the object in the global frame?



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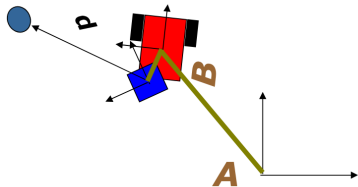


Bp gives the pose of the object wrt the robot

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Bp gives the pose of the object wrt the robot

ABp gives the pose of the object wrt the world